

```

Last login: Thu Jun 18 13:25:13 on ttys000
stephenrajva@Stephens-MacBook-Air-4 ~ % cd ~/Downloads/SST_Stimulator/MiniFE
stephenrajva@Stephens-MacBook-Air-4 MiniFE % pwd
/Users/stephenrajva/Downloads/SST_Stimulator/MiniFE
stephenrajva@Stephens-MacBook-Air-4 MiniFE % ls
DragonFly      Torus
stephenrajva@Stephens-MacBook-Air-4 MiniFE % cd ~/Downloads/SST_Stimulator/MiniFE/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % pwd
/Users/stephenrajva/Downloads/SST_Stimulator/MiniFE/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -la
total 0
drwxr-xr-x  2 stephenrajva  staff   64 18 Jun 13:25 .
drwxr-xr-x  5 stephenrajva  staff  160 18 Jun 13:25 ..
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniFE/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % nano torus4.py
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -l
total 8
-rw-r--r--  1 stephenrajva  staff  2538 18 Jun 13:44 torus4.py
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniFE/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % sst torus4.py
Allreduce: ranks 4, loop 92, 1 double(s), latency 3.830 us
generate
355809533:0:0:EmberEngine:Reduce:generate:37: rank=0 size=4
generate
356609133:0:2:EmberEngine:Reduce:generate:37: rank=2 size=4
generate
356849133:0:1:EmberEngine:Reduce:generate:37: rank=1 size=4
generate
Bcast: ranks 4, loop 2, bytes 56, latency 1.763 us
357648333:0:3:EmberEngine:Reduce:generate:37: rank=3 size=4
Allgather: ranks 4, loop 3, bytes 12, latency 4.061 us
Barrier: ranks 4, loop 99, latency 3.829 us
773781133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=1 messageSize=409 iterations=35
774580333:0:2:EmberEngine:PingPong:generate:65: rank=2 size=4
774580733:0:1:EmberEngine:PingPong:generate:65: rank=1 size=4
775380333:0:3:EmberEngine:PingPong:generate:65: rank=3 size=4
PingPong: otherRank 1, total time 69.392 us, loop 35, bufLen 409, latency 0.991 us. bandwidth 0.412584 G
B/s
843173133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=2 messageSize=409 iterations=35
911645533:0:2:EmberEngine:PingPong:generate:65: rank=2 size=4
PingPong: otherRank 2, total time 69.272 us, loop 35, bufLen 409, latency 0.990 us. bandwidth 0.413298 G
B/s
912445133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=3 messageSize=82 iterations=35
986879133:0:3:EmberEngine:PingPong:generate:65: rank=3 size=4
PingPong: otherRank 3, total time 75.320 us, loop 35, bufLen 82, latency 1.076 us. bandwidth 0.076208 GB
/s
987765133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=2 messageSize=82 iterations=35
PingPong: otherRank 2, total time 69.272 us, loop 35, bufLen 82, latency 0.990 us. bandwidth 0.082862 GB
/s
1057037133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=3 messageSize=490 iterations=35
1131471133:0:3:EmberEngine:PingPong:generate:65: rank=3 size=4
PingPong: otherRank 3, total time 75.320 us, loop 35, bufLen 490, latency 1.076 us. bandwidth 0.455390 G
B/s
1132357133:0:0:EmberEngine:PingPong:generate:65: rank=0 size=4
rank2=3 messageSize=490 iterations=35
PingPong: otherRank 3, total time 75.320 us, loop 35, bufLen 490, latency 1.076 us. bandwidth 0.455390 G
B/s
Simulation is complete, simulated time: 1.21057 ms
stephenrajva@Stephens-MacBook-Air-4 Torus % ls *.csv
miniFE_communication_stats_torus.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "send_bit_count" miniFE_communication_stats_torus.csv |
head
rtr_0x0, send_bit_count, port0, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port1, Accumulator, 1210569133, 0, 541352, 1811934528, 340, 360, 4280
rtr_0x0, send_bit_count, port2, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port3, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port4, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port5, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port6, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port7, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port8, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0

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rtr_0x0, send_bit_count, port9, Accumulator, 1210569133, 0, 243408, 530560640, 273, 360, 3632
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "send_packet_count" miniFE_communication_stats_torus.csv | head
rtr_0x0, send_packet_count, port0, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port1, Accumulator, 1210569133, 0, 340, 340, 340, 1, 1
rtr_0x0, send_packet_count, port2, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port3, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port4, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port5, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port6, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port7, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port8, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, send_packet_count, port9, Accumulator, 1210569133, 0, 273, 273, 273, 1, 1
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "output_port_stalls" miniFE_communication_stats_torus.csv | head
rtr_0x0, output_port_stalls, port0, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port1, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port2, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port3, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port4, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port5, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port6, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port7, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port8, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
rtr_0x0, output_port_stalls, port9, Accumulator, 1210569133, 0, 0, 0, 0, 0, 0
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "idle_time" miniFE_communication_stats_torus.csv | head
rtr_0x0, idle_time, port0, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port1, Accumulator, 1210569133, 0, 1204733259, 14403570956918349, 341, 126401, 71413
336
rtr_0x0, idle_time, port2, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port3, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port4, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port5, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port6, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port7, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port8, Accumulator, 1210569133, 0, 1210569133, 1465477625772371689, 1, 1210569133, 1
210569133
rtr_0x0, idle_time, port9, Accumulator, 1210569133, 0, 1207885804, 38214601426495144, 274, 125068, 15371
0936
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "packet_latency" miniFE_communication_stats_torus.csv | head
nic0:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 129941, 28499721, 607, 198, 285
nic1:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 85625, 16971461, 432, 198, 199
nic2:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 54550, 10943882, 273, 197, 285
nic3:rtrLink:networkIF, packet_latency, , Accumulator, 1210569133, 0, 70158, 16502522, 308, 197, 285
stephenrajva@Stephens-MacBook-Air-4 Torus %
stephenrajva@Stephens-MacBook-Air-4 Torus % cd ~/Downloads/SST_Stimulator/MiniFE/Torus
stephenrajva@Stephens-MacBook-Air-4 Torus % cp torus4.py torus16.py
stephenrajva@Stephens-MacBook-Air-4 Torus % nano torus16.py
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "shape" torus16.py
topo.shape = "4x4"
stephenrajva@Stephens-MacBook-Air-4 Torus % grep '"size"' torus16.py
"size": 16,
ep = EmberMPIJob(0, job["size"])
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "filepath" torus16.py
"filepath": "miniFE_communication_stats_torus.csv",
stephenrajva@Stephens-MacBook-Air-4 Torus % nano torus16.py
stephenrajva@Stephens-MacBook-Air-4 Torus % rm -f miniFE_communication_stats_torus16.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % sst torus16.py
File "/Users/stephenrajva/Downloads/SST_Stimulator/MiniFE/Torus/torus16.py", line 75
    separator": ", "
                    ^
SyntaxError: unterminated string literal (detected at line 75)
FATAL: # SSTPythonModel: Execution of model construction function failed
SST Fatal Backtrace Information:
0 : 0 sstsim.x 0x0000000100a093c0 _ZNK3SST6Output5fatalEjPKcS2_is2_z +
644
1 : 1 sstsim.x 0x0000000100ba70c4 _ZN3SST4Core24SSTPythonModelDefinitio
n17createConfigGraphEv + 504

```

```
2 : 2 sstsim.x 0x0000000100969524 main + 2692
3 : 3 dyld 0x0000000186211d54 start + 7184
```

```
-----
MPI_ABORT was invoked on rank 0 in communicator MPI_COMM_WORLD
Proc: [[8586,0],0]
Errorcode: 1
```

NOTE: invoking MPI_ABORT causes Open MPI to kill all MPI processes.
You may or may not see output from other processes, depending on
exactly when Open MPI kills them.

```
-----
stephenrajva@Stephens-MacBook-Air-4 Torus % nl -ba torus16.py | sed -n '68,80p'
```

```
68
69 system.build()
70
71 sst.setStatisticLoadLevel(10)
72
73 sst.setStatisticOutput("sst.statOutputCSV", {
74     "filepath": "miniFE_communication_stats_torus16.csv",
75     "separator": ",",
76 })
77
78 sst.enableAllStatisticsForComponentType("merlin.hr_router")
79 sst.enableAllStatisticsForComponentType("merlin.linkcontrol")
80
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % nano torus16.py
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % nl -ba torus16.py | sed -n '73,76p'
```

```
73 sst.setStatisticOutput("sst.statOutputCSV", {
74     "filepath": "miniFE_communication_stats_torus16.csv",
75     "separator": ",",
76 })
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % rm -f miniFE_communication_stats_torus16.csv
```

```
stephenrajva@Stephens-MacBook-Air-4 Torus % sst torus16.py
```

```
Allreduce: ranks 16, loop 92, 1 double(s), latency 7.875 us
```

```
generate
```

```
727935933:0:0:EmberEngine:Reduce:generate:37: rank=0 size=16
```

```
generate
```

```
729365533:0:1:EmberEngine:Reduce:generate:37: rank=1 size=16
```

```
generate
```

```
729581533:0:2:EmberEngine:Reduce:generate:37: rank=2 size=16
```

```
generate
```

```
730381533:0:3:EmberEngine:Reduce:generate:37: rank=3 size=16
```

```
generate
```

```
730511733:0:4:EmberEngine:Reduce:generate:37: rank=4 size=16
```

```
generate
```

```
730597533:0:5:EmberEngine:Reduce:generate:37: rank=5 size=16
```

```
generate
```

```
730641333:0:6:EmberEngine:Reduce:generate:37: rank=6 size=16
```

```
generate
```

```
731180733:0:7:EmberEngine:Reduce:generate:37: rank=7 size=16
```

```
generate
```

```
731267533:0:9:EmberEngine:Reduce:generate:37: rank=9 size=16
```

```
generate
```

```
731353933:0:8:EmberEngine:Reduce:generate:37: rank=8 size=16
```

```
generate
```

```
731439533:0:11:EmberEngine:Reduce:generate:37: rank=11 size=16
```

```
generate
```

```
731483533:0:13:EmberEngine:Reduce:generate:37: rank=13 size=16
```

```
generate
```

```
731484333:0:10:EmberEngine:Reduce:generate:37: rank=10 size=16
```

```
generate
```

```
731527533:0:14:EmberEngine:Reduce:generate:37: rank=14 size=16
```

```
generate
```

```
731569933:0:12:EmberEngine:Reduce:generate:37: rank=12 size=16
```

```
generate
```

```
Bcast: ranks 16, loop 2, bytes 56, latency 1.893 us
```

```
732066733:0:15:EmberEngine:Reduce:generate:37: rank=15 size=16
```

```
Allgather: ranks 16, loop 3, bytes 12, latency 6.480 us
```

```
Barrier: ranks 16, loop 99, latency 7.872 us
```

```
1558541933:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16  
rank2=1 messageSize=409 iterations=35
```

```
1559471533:0:1:EmberEngine:PingPong:generate:65: rank=1 size=16
```

```
1559687533:0:2:EmberEngine:PingPong:generate:65: rank=2 size=16
```

```
1560487533:0:3:EmberEngine:PingPong:generate:65: rank=3 size=16
```

```
PingPong: otherRank 1, total time 69.522 us, loop 35, bufLen 409, latency 0.993 us. bandwidth 0.411812 G  
B/s
```

```

1628063533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=2 messageSize=409 iterations=35
1702497533:0:2:EmberEngine:PingPong:generate:65: rank=2 size=16
PingPong: otherRank 2, total time 75.320 us, loop 35, bufLen 409, latency 1.076 us. bandwidth 0.380112 G
B/s
1703383533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=3 messageSize=82 iterations=35
1771855933:0:3:EmberEngine:PingPong:generate:65: rank=3 size=16
PingPong: otherRank 3, total time 69.272 us, loop 35, bufLen 82, latency 0.990 us. bandwidth 0.082862 GB
/s
1772655533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=2 messageSize=82 iterations=35
PingPong: otherRank 2, total time 75.320 us, loop 35, bufLen 82, latency 1.076 us. bandwidth 0.076208 GB
/s
1847975533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=3 messageSize=490 iterations=35
1916447933:0:3:EmberEngine:PingPong:generate:65: rank=3 size=16
PingPong: otherRank 3, total time 69.272 us, loop 35, bufLen 490, latency 0.990 us. bandwidth 0.495150 G
B/s
1917247533:0:0:EmberEngine:PingPong:generate:65: rank=0 size=16
rank2=3 messageSize=490 iterations=35
PingPong: otherRank 3, total time 69.272 us, loop 35, bufLen 490, latency 0.990 us. bandwidth 0.495150 G
B/s
Simulation is complete, simulated time: 1.99227 ms
stephenrajva@Stephens-MacBook-Air-4 Torus % ls *.csv
miniFE_communication_stats_torus.csv      miniFE_communication_stats_torus16.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "send_bit_count" miniFE_communication_stats_torus16.csv
| head
rtr_0x0, send_bit_count, port0, Accumulator, 1992266733, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port1, Accumulator, 1992266733, 0, 283832, 524407360, 432, 360, 3632
rtr_0x0, send_bit_count, port2, Accumulator, 1992266733, 0, 240960, 529548032, 267, 360, 3632
rtr_0x0, send_bit_count, port3, Accumulator, 1992266733, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port4, Accumulator, 1992266733, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port5, Accumulator, 1992266733, 0, 335160, 1318416960, 105, 1016, 4280
rtr_0x0, send_bit_count, port6, Accumulator, 1992266733, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port7, Accumulator, 1992266733, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port8, Accumulator, 1992266733, 0, 0, 0, 0, 0, 0
rtr_0x0, send_bit_count, port9, Accumulator, 1992266733, 0, 86344, 38209600, 209, 360, 1128
stephenrajva@Stephens-MacBook-Air-4 Torus % grep "packet_latency" miniFE_communication_stats_torus16.csv
| head
nic0:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 145552, 35735396, 613, 198, 371
nic1:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 159569, 41436407, 632, 198, 371
nic2:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 172159, 45483633, 667, 197, 285
nic3:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 189937, 54999955, 702, 198, 372
nic4:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 186597, 59349297, 597, 198, 372
nic5:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 202916, 70131966, 597, 197, 376
nic6:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 169058, 50849830, 597, 197, 372
nic7:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 97151, 24172237, 403, 198, 285
nic8:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 76166, 27905496, 209, 198, 372
nic9:rtrLink:networkIF, packet_latency, , Accumulator, 1992266733, 0, 58961, 16721789, 209, 198, 371
stephenrajva@Stephens-MacBook-Air-4 Torus %
stephenrajva@Stephens-MacBook-Air-4 Torus % cp torus16.py torus64.py
stephenrajva@Stephens-MacBook-Air-4 Torus % nano torus64.py
stephenrajva@Stephens-MacBook-Air-4 Torus % sst torus64.py
Allreduce: ranks 64, loop 92, 1 double(s), latency 13.015 us
generate
1200874333:0:0:EmberEngine:Reduce:generate:37: rank=0 size=64
generate
1202303933:0:1:EmberEngine:Reduce:generate:37: rank=1 size=64
generate
1202519933:0:2:EmberEngine:Reduce:generate:37: rank=2 size=64
generate
1203319933:0:3:EmberEngine:Reduce:generate:37: rank=3 size=64
generate
1203536533:0:4:EmberEngine:Reduce:generate:37: rank=4 size=64
generate
1203622333:0:5:EmberEngine:Reduce:generate:37: rank=5 size=64
generate
1203838933:0:6:EmberEngine:Reduce:generate:37: rank=6 size=64
generate
1204508733:0:7:EmberEngine:Reduce:generate:37: rank=7 size=64
generate
1204638933:0:8:EmberEngine:Reduce:generate:37: rank=8 size=64
generate
1204724733:0:11:EmberEngine:Reduce:generate:37: rank=11 size=64
generate

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1204725333:0:9:EmberEngine:Reduce:generate:37: rank=9 size=64
generate
1204768533:0:12:EmberEngine:Reduce:generate:37: rank=12 size=64
generate
1204769133:0:10:EmberEngine:Reduce:generate:37: rank=10 size=64
generate
1204854933:0:13:EmberEngine:Reduce:generate:37: rank=13 size=64
generate
1204898733:0:14:EmberEngine:Reduce:generate:37: rank=14 size=64
generate
1205438333:0:15:EmberEngine:Reduce:generate:37: rank=15 size=64
generate
1205654933:0:17:EmberEngine:Reduce:generate:37: rank=17 size=64
generate
1205741333:0:16:EmberEngine:Reduce:generate:37: rank=16 size=64
generate
1205827733:0:19:EmberEngine:Reduce:generate:37: rank=19 size=64
generate
1205871533:0:18:EmberEngine:Reduce:generate:37: rank=18 size=64
generate
1205957933:0:21:EmberEngine:Reduce:generate:37: rank=21 size=64
generate
1205999933:0:23:EmberEngine:Reduce:generate:37: rank=23 size=64
generate
1206001133:0:29:EmberEngine:Reduce:generate:37: rank=29 size=64
generate
1206043733:0:25:EmberEngine:Reduce:generate:37: rank=25 size=64
generate
1206043733:0:27:EmberEngine:Reduce:generate:37: rank=27 size=64
generate
1206044333:0:20:EmberEngine:Reduce:generate:37: rank=20 size=64
generate
1206044333:0:30:EmberEngine:Reduce:generate:37: rank=30 size=64
generate
1206087533:0:26:EmberEngine:Reduce:generate:37: rank=26 size=64
generate
1206087533:0:28:EmberEngine:Reduce:generate:37: rank=28 size=64
generate
1206130133:0:24:EmberEngine:Reduce:generate:37: rank=24 size=64
generate
1206173933:0:22:EmberEngine:Reduce:generate:37: rank=22 size=64
generate
1206323933:0:31:EmberEngine:Reduce:generate:37: rank=31 size=64
generate
1206497133:0:32:EmberEngine:Reduce:generate:37: rank=32 size=64
generate
1206583533:0:33:EmberEngine:Reduce:generate:37: rank=33 size=64
generate
1206583533:0:35:EmberEngine:Reduce:generate:37: rank=35 size=64
generate
1206800333:0:34:EmberEngine:Reduce:generate:37: rank=34 size=64
generate
1206800333:0:36:EmberEngine:Reduce:generate:37: rank=36 size=64
generate
1206841933:0:47:EmberEngine:Reduce:generate:37: rank=47 size=64
generate
1206886733:0:37:EmberEngine:Reduce:generate:37: rank=37 size=64
generate
1206929133:0:39:EmberEngine:Reduce:generate:37: rank=39 size=64
generate
1206973133:0:43:EmberEngine:Reduce:generate:37: rank=43 size=64
generate
1207016333:0:44:EmberEngine:Reduce:generate:37: rank=44 size=64
generate
1207058733:0:49:EmberEngine:Reduce:generate:37: rank=49 size=64
generate
1207058733:0:51:EmberEngine:Reduce:generate:37: rank=51 size=64
generate
1207059533:0:40:EmberEngine:Reduce:generate:37: rank=40 size=64
generate
1207059533:0:61:EmberEngine:Reduce:generate:37: rank=61 size=64
generate
1207102733:0:38:EmberEngine:Reduce:generate:37: rank=38 size=64
generate
1207102733:0:45:EmberEngine:Reduce:generate:37: rank=45 size=64
generate

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1207102733:0:59:EmberEngine:Reduce:generate:37: rank=59 size=64
generate
1207102733:0:62:EmberEngine:Reduce:generate:37: rank=62 size=64
generate
1207145133:0:48:EmberEngine:Reduce:generate:37: rank=48 size=64
generate
1207145933:0:41:EmberEngine:Reduce:generate:37: rank=41 size=64
generate
1207145933:0:46:EmberEngine:Reduce:generate:37: rank=46 size=64
generate
1207145933:0:60:EmberEngine:Reduce:generate:37: rank=60 size=64
generate
1207189133:0:42:EmberEngine:Reduce:generate:37: rank=42 size=64
generate
1207189133:0:53:EmberEngine:Reduce:generate:37: rank=53 size=64
generate
1207231533:0:55:EmberEngine:Reduce:generate:37: rank=55 size=64
generate
1207275533:0:50:EmberEngine:Reduce:generate:37: rank=50 size=64
generate
1207275533:0:52:EmberEngine:Reduce:generate:37: rank=52 size=64
generate
1207275533:0:57:EmberEngine:Reduce:generate:37: rank=57 size=64
generate
1207318733:0:58:EmberEngine:Reduce:generate:37: rank=58 size=64
generate
1207361933:0:56:EmberEngine:Reduce:generate:37: rank=56 size=64
generate
Bcast: ranks 64, loop 2, bytes 56, latency 1.893 us
1207382733:0:63:EmberEngine:Reduce:generate:37: rank=63 size=64
generate
1207405133:0:54:EmberEngine:Reduce:generate:37: rank=54 size=64
Allgather: ranks 64, loop 3, bytes 12, latency 9.377 us
Barrier: ranks 64, loop 99, latency 13.009 us
2554822733:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=1 messageSize=409 iterations=35
2555752333:0:1:EmberEngine:PingPong:generate:65: rank=1 size=64
2555968333:0:2:EmberEngine:PingPong:generate:65: rank=2 size=64
2556768333:0:3:EmberEngine:PingPong:generate:65: rank=3 size=64
PingPong: otherRank 1, total time 69.522 us, loop 35, bufLen 409, latency 0.993 us. bandwidth 0.411812 G
B/s
2624344333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=2 messageSize=409 iterations=35
2698778333:0:2:EmberEngine:PingPong:generate:65: rank=2 size=64
PingPong: otherRank 2, total time 75.320 us, loop 35, bufLen 409, latency 1.076 us. bandwidth 0.380112 G
B/s
2699664333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=3 messageSize=82 iterations=35
2780059933:0:3:EmberEngine:PingPong:generate:65: rank=3 size=64
PingPong: otherRank 3, total time 81.368 us, loop 35, bufLen 82, latency 1.162 us. bandwidth 0.070544 GB
/s
2781032333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=2 messageSize=82 iterations=35
PingPong: otherRank 2, total time 75.320 us, loop 35, bufLen 82, latency 1.076 us. bandwidth 0.076208 GB
/s
2856352333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=3 messageSize=490 iterations=35
2936747933:0:3:EmberEngine:PingPong:generate:65: rank=3 size=64
PingPong: otherRank 3, total time 81.368 us, loop 35, bufLen 490, latency 1.162 us. bandwidth 0.421542 G
B/s
2937720333:0:0:EmberEngine:PingPong:generate:65: rank=0 size=64
rank2=3 messageSize=490 iterations=35
PingPong: otherRank 3, total time 81.368 us, loop 35, bufLen 490, latency 1.162 us. bandwidth 0.421542 G
B/s
Simulation is complete, simulated time: 3.02706 ms
stephenrajva@Stephens-MacBook-Air-4 Torus %
stephenrajva@Stephens-MacBook-Air-4 Torus % ls *.csv
miniFE_communication_stats_torus.csv miniFE_communication_stats_torus64.csv
miniFE_communication_stats_torus16.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % nano extract_torus_metrics.py
stephenrajva@Stephens-MacBook-Air-4 Torus % python3 extract_torus_metrics.py
Topology      SendBits  Packets  Stalls  AvgLatency  AvgIdleTime  Throughput
0  Torus4    16876680576  3462    0  263.500000  9.950634e+08  6.404812e+07
1  Torus16   18874219392  19840    0  350.125000  1.461280e+09  5.390709e+07
2  Torus64   61481485952  130364    0  561.984375  1.604725e+09  1.094007e+08

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Saved: torus_summary.csv
stephenrajva@Stephens-MacBook-Air-4 Torus %
stephenrajva@Stephens-MacBook-Air-4 Torus % ls -lh *.csv
-rw-r--r--  1 stephenrajva  staff    35K 18 Jun 13:44 miniFE_communication_stats_torus.csv
-rw-r--r--  1 stephenrajva  staff   139K 18 Jun 20:41 miniFE_communication_stats_torus16.csv
-rw-r--r--  1 stephenrajva  staff   560K 18 Jun 20:50 miniFE_communication_stats_torus64.csv
-rw-r--r--  1 stephenrajva  staff    284B 18 Jun 20:51 torus_summary.csv
stephenrajva@Stephens-MacBook-Air-4 Torus % mkdir -p ~/Downloads/SST_Stimulator/MiniFE/Comparison

cd ~/Downloads/SST_Stimulator/MiniFE/Comparison

pwd
/Users/stephenrajva/Downloads/SST_Stimulator/MiniFE/Comparison
stephenrajva@Stephens-MacBook-Air-4 Comparison % cp ~/Downloads/SST_Stimulator/MiniFE/DragonFly/dragonfly_summary.csv .
stephenrajva@Stephens-MacBook-Air-4 Comparison % cp ~/Downloads/SST_Stimulator/MiniFE/Torus/torus_summary.csv .
stephenrajva@Stephens-MacBook-Air-4 Comparison % ls -lh
total 16
-rw-r--r--  1 stephenrajva  staff    220B 18 Jun 20:54 dragonfly_summary.csv
-rw-r--r--  1 stephenrajva  staff    284B 18 Jun 20:54 torus_summary.csv
stephenrajva@Stephens-MacBook-Air-4 Comparison % ls -lh
total 16
-rw-r--r--  1 stephenrajva  staff    220B 18 Jun 20:54 dragonfly_summary.csv
-rw-r--r--  1 stephenrajva  staff    284B 18 Jun 20:54 torus_summary.csv
stephenrajva@Stephens-MacBook-Air-4 Comparison % nano combine_results.py
stephenrajva@Stephens-MacBook-Air-4 Comparison % python3 combine_results.py

===== COMBINED DATA =====
  Topology    SendBits  Packets  Stalls  AvgLatency    AvgIdleTime    Throughput    Family
0   DFLy4      5847696    3462     0    263.500000    2.408114e+08    2.219239e+04    DragonFly
1   DFLy16    14066216    20345     0    366.500000    6.622161e+08    3.837985e+04    DragonFly
2   DFLy64    48972184    80627     0    383.380000    4.777996e+08    1.277396e+05    DragonFly
3   Torus4    16876680576    3462     0    263.500000    9.950634e+08    6.404812e+07    Torus
4   Torus16   18874219392    19840     0    350.125000    1.461280e+09    5.390709e+07    Torus
5   Torus64   61481485952   130364     0    561.984375    1.604725e+09    1.094007e+08    Torus

Saved: combined_topology_comparison.csv
stephenrajva@Stephens-MacBook-Air-4 Comparison % nano plot_topology_comparison.py
stephenrajva@Stephens-MacBook-Air-4 Comparison % python3 plot_topology_comparison.py
Created:
latency_comparison.png
packet_comparison.png
idletime_comparison.png
stephenrajva@Stephens-MacBook-Air-4 Comparison % open latency_comparison.png
open packet_comparison.png
open idletime_comparison.png
stephenrajva@Stephens-MacBook-Air-4 Comparison %

```